ADDISON DOBBS

907-315-8399 • addisontdobbs@gmail.com • linkedin.com/in/addisontdobbs • addisontdobbs.github.io

SUMMARY

Computer Science graduate at the University of Alaska Anchorage focused on software engineering and data analysis. Experience building full-stack web tools for research, with a strong emphasis on database design, data visualization, and maintainable systems.

EXPERIENCE

Research Assistant - University of Alaska Anchorage **Feb 2026 - Present**

- Building a **full-stack** web application for earth science field-data collection and visualization using **Django** and **PostgreSQL**.
- Implementing interactive mapping with **Leaflet.js** and data charts with **Plotly.js** to help researchers explore visual maps and field data.
- Designing structured forms and data models to streamline field workflows and keep collected data consistent.

PROJECTS

Alaska Aviation Data Explorer **August 2026 - Present**

- Developing a **full-stack** Alaska Aviation Data Explorer for analyzing flight performance and aviation trends.
- Cleaning, validating, and organizing **64,000+** U.S. DOT flight records, using **SQL** to analyze routes, carriers, delays, cancellations, and operational trends.
- Creating a **Django**-based web application with an interactive dashboard that connects **database** results to reports, summaries, and visualizations for easier data analysis.

AI Sudoku Solver

- Programmed a Sudoku solver and puzzle generator using **Python**, comparing standard backtracking with **Minimum Remaining Values** and **forward checking**.
- Implemented uniqueness validation and tested solver performance across multiple puzzles to evaluate runtime and search efficiency.

Portfolio Website

- Created a fully functional personal website using **HTML**, **CSS**, and **JavaScript** featuring interactive **UI** elements and custom animations.
- Designed responsive layouts with scroll-triggered reveals, dark/light theme, project modals, and a reading progress bar.

EDUCATION

Bachelor of Science in Computer Science **Expected December 2026**

- Minor in Computer Systems Engineering

University of Alaska Anchorage

ADDITIONAL INFORMATION

- **Technical Skills:** HTML/CSS, JavaScript, Python, C++, SQL
- **Core Concepts:** Object-Oriented Programming, Visualization, Debugging, Problem Solving
- **Professional Skills:** Adaptability, Time Management, Written and Oral Communication, Attention to Detail